

Optical Distance Calculation for the Projector Mirror Conical Screen Setup

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Goal

We want the horizontal projector mirror distance and the vertical mirror screen distance such that the projected coordinates produced by the transformation

$$s = \frac{1}{\alpha r + \beta z}, \quad \alpha = 1.7, \quad \beta = -1.2,$$

are displayed on the conical screen with stable geometry and without additional angular distortion from the folded optical path.

1. Physical screen geometry

The conical screen is specified by:

$$D_{\max} \approx 220 \text{ mm}, \quad D_{\min} \approx 40 \text{ mm}, \quad H \approx 60 \text{ mm}.$$

Hence the corresponding radii are

$$R_{\max} = \frac{D_{\max}}{2} = 110 \text{ mm}, \quad R_{\min} = \frac{D_{\min}}{2} = 20 \text{ mm}.$$

If we measure vertical position on the screen by $z \in [0, H]$, from the small opening to the large opening, then the physical cone radius is

$$r_{\text{phys}}(z) = R_{\min} + \frac{R_{\max} - R_{\min}}{H} z = 20 + \frac{110 - 20}{60} z = 20 + 1.5z.$$

So the physical cone slope is

$$\frac{dr_{\text{phys}}}{dz} = 1.5,$$

and the corresponding cone half-angle (measured from the axis) is

$$\theta_{\text{phys}} = \tan^{-1}(1.5) \approx 56.3^\circ.$$

2. Geometry implied by the transformation

The transformation is

$$r = \sqrt{x^2 + y^2}, \quad s = \frac{1}{\alpha r + \beta z},$$

and the normalized displayed radius is

$$\rho = rs = \frac{r}{\alpha r + \beta z}.$$

The screen edge corresponds to $\rho = 1$. Therefore

$$\frac{r}{\alpha r + \beta z} = 1 \implies r = \alpha r + \beta z.$$

Rearranging,

$$(1 - \alpha)r = \beta z.$$

Substituting $\alpha = 1.7$ and $\beta = -1.2$,

$$(1 - 1.7)r = -1.2z \implies -0.7r = -1.2z,$$

hence

$$z = \frac{0.7}{1.2}r \approx 0.5833 r,$$

or equivalently

$$r \approx 1.714 z.$$

Thus the cone half-angle implied by the transformation is

$$\theta_{\text{model}} = \tan^{-1}\left(\frac{r}{z}\right) = \tan^{-1}(1.714) \approx 59.7^\circ.$$

Interpretation

The transformation therefore encodes a cone with half-angle about

$$\theta_{\text{model}} \approx 59.7^\circ,$$

while the physical screen has

$$\theta_{\text{phys}} \approx 56.3^\circ.$$

These are close, so the transformation is consistent with the stated conical screen geometry up to the empirical corrections already present in the code:

$$\text{LRShift} = -0.15, \quad \text{APShift} = -0.25.$$

Therefore the transformation is mainly responsible for the *angular remapping* onto the cone, while the projector mirror screen distances are determined by the *optical throw geometry*.

3. Optical path constraint

Let

- x = horizontal distance from projector beam exit to the mirror center,
- y = vertical distance from the mirror center to the *mid-height* of the active conical screen,
- L = total folded optical path.

With a 45° mirror, the path is folded by 90° , so

$$L = x + y.$$

Now let T be the projector throw ratio. If the projected field must cover the full usable width of the screen, then

$$L = T D_{\max}.$$

Since $D_{\max} = 220$ mm,

$$L = 220 T.$$

Therefore the fundamental constraint is

$$\boxed{x + y = 220 T.}$$

4. Numerical result under the standard DLP3010 assumption

If we take the commonly used assumption

$$T = 1.2,$$

then

$$L = 220 \times 1.2 = 264 \text{ mm.}$$

Hence

$$\boxed{x + y = 264 \text{ mm.}}$$

This is the *correct total folded optical path*. The transformation itself does *not* uniquely determine the split between x and y ; it determines the cone remapping, while only the projector optics determine the total path.

5. Symmetric design split

A practical stable choice is to keep the two legs nearly balanced:

$$x \approx 130 \text{ mm}, \quad y \approx 134 \text{ mm},$$

since

$$130 + 134 = 264.$$

Therefore the recommended optical distances are

$$\boxed{x \approx 130 \text{ mm}}$$

for the projector beam exit to the mirror center, and

$$\boxed{y \approx 134 \text{ mm}}$$

for the mirror center to the *screen mid-height*.

6. If the screen is measured from the top rim instead of the mid-height

Because the cone height is

$$H = 60 \text{ mm},$$

the top rim is 30 mm above the screen mid-height. Therefore:

$$y_{\text{top}} = y + \frac{H}{2} = 134 + 30 = 164 \text{ mm},$$

and the bottom rim is

$$y_{\text{bottom}} = y - \frac{H}{2} = 134 - 30 = 104 \text{ mm}.$$

So, depending on how the screen distance is measured:

mirror center to screen mid-height $\approx 134 \text{ mm}$, mirror center to top rim $\approx 164 \text{ mm}$, mirror center to bottom rim $\approx 104 \text{ mm}$.
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7. Final result

The transformation confirms that the screen geometry is conical and close to the physical cone dimensions. The optical distances are then set by the projector throw condition:

$$x + y = 220 T.$$

Under the standard assumption $T = 1.2$,

$$x + y = 264 \text{ mm}.$$

A stable and practical split is

$x \approx 130 \text{ mm}, \quad y \approx 134 \text{ mm}.$

If the vertical measurement is taken to the top of the cone instead of the cone mid-height, then use

$y_{\text{top}} \approx 164 \text{ mm}.$
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Realistic Approach

Realistic interpretation of the distances

The distance

$$x$$

is the horizontal projector–mirror distance, that is, the distance traveled by the beam before reflection. Physically, this distance controls how the beam reaches the mirror and how sensitive the setup is to small alignment errors. If x is too small, small angular errors at the mirror produce larger geometric changes on the screen. If x is too large, the setup becomes unnecessarily extended without improving the cone mapping.

The distance

$$y$$

is the vertical mirror–screen distance after reflection. This is the effective projection distance after the fold and is the quantity most directly responsible for the final image size on the screen.

The key point is that the projector optics depend on the *total optical path*, not on the fact that a mirror is present. Therefore the relevant optical constraint is

$$x + y = L,$$

where L is the total throw distance required to project the intended image width.

Thus:

- x mainly affects beam conditioning and alignment sensitivity,
- y mainly affects the actual projected image size,
- $x + y$ is the optical quantity constrained by the projector throw ratio.

In practice, this means that the transformation determines the angular remapping onto the cone, while the distances ensure that the projected field has the correct overall size.

Realistic interpretation of the transformation

The transformation

$$s = \frac{1}{\alpha r + \beta z}, \quad \alpha = 1.7, \quad \beta = -1.2,$$

does not itself determine the optical path lengths. Instead, it determines how points in the virtual stimulus are remapped onto the conical screen.

This means that the transformation is responsible for correcting the geometry of the projected stimulus so that a point displayed by the projector appears at the intended angular position on the cone.

Hence, in a realistic interpretation:

- the transformation corrects for the conical screen geometry,
- the optical distances are chosen so that the projected field physically covers the intended part of the cone.

Because the transformation-implied cone angle is close to the physical cone angle, the mapping is already approximately matched to the hardware. Therefore the optical distances should not be treated as geometric correction parameters, but rather as projection-scale parameters.

Realistic choice of the distance split

From the projector throw condition,

$$x + y = 220 T.$$

If $T = 1.2$, then

$$x + y = 264 \text{ mm}.$$

This total path is fixed by the optics, but the split between x and y is not unique. A realistic strategy is to avoid extreme values:

$$x \rightarrow 0, \quad y \rightarrow 264$$

would place the mirror too close to the projector, making alignment highly sensitive, whereas

$$x \rightarrow 264, \quad y \rightarrow 0$$

would place the image-forming region too close to the mirror, which is impractical for a conical screen.

A balanced split such as

$$x \approx 130 \text{ mm}, \quad y \approx 134 \text{ mm}$$

is therefore realistic because it gives:

- a stable pre-reflection beam path,
- a sufficient post-reflection imaging distance,
- a symmetric folded geometry,
- reduced sensitivity to small angular misalignments.

Realistic mirror-angle choice

In the ideal folded geometry, the beam travels horizontally before reflection and vertically after reflection. This requires a direction change of

$$90^\circ.$$

For a planar mirror, the reflected direction is obtained by symmetry about the mirror normal, so the mirror normal bisects the angle between incident and reflected rays.

Thus the required mirror angle is

$$\theta_{\text{mirror}} = 45^\circ.$$

This is the correct *design angle*. In practice, however, real hardware introduces small deviations:

- the projector may not be perfectly level,
- the mirror mount may not be perfectly centered,
- the screen center may not be exactly above the mirror center,
- machining tolerances may shift the real geometry.

Therefore, in a realistic setup, the best strategy is:

- use

$$45^\circ$$

as the nominal design angle,

- allow a small adjustment around this value during alignment.

A typical realistic tuning range is

$$44^\circ \text{ to } 46^\circ,$$

or slightly wider if needed. This adjustment is not used to redefine the cone mapping, but only to compensate for small hardware offsets.

Hence the realistic conclusion is:

$\theta_{\text{mirror}} \approx 45^\circ$ nominal, with small fine adjustment around it.
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Realistic measurement convention

In practice, the mirror–screen distance must be defined relative to a specific reference point on the cone. The mathematically most natural reference is the screen mid-height. However, experimentally one may instead measure to the top rim or bottom rim.

If y denotes the mirror center to screen mid-height distance, then

$$y_{\text{top}} = y + \frac{H}{2}, \quad y_{\text{bottom}} = y - \frac{H}{2}.$$

For

$$H = 60 \text{ mm}, \quad y = 134 \text{ mm},$$

this gives

$$y_{\text{top}} = 164 \text{ mm}, \quad y_{\text{bottom}} = 104 \text{ mm}.$$

Thus, in real setup work, it is important to specify whether the vertical distance is measured to:

- the mid-height of the cone,
- the top rim,
- or the bottom rim.

Realistic practical conclusion

The realistic interpretation of the system is therefore:

1. The transformation sets the angular remapping required by the cone.
2. The projector throw determines the total optical path:

$$x + y = 220 T.$$

3. The mirror folds the path but does not change the optical throw requirement.
4. A balanced split of the total path is preferable for alignment stability.
5. The mirror angle should be designed at

$$45^\circ$$

and only fine-tuned slightly in practice.

Under the standard assumption $T = 1.2$, the realistic recommended distances remain

$x \approx 130 \text{ mm}, \quad y \approx 134 \text{ mm}, \quad \theta_{\text{mirror}} \approx 45^\circ.$
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Ring Design for Conical Screen Mounting

Given screen geometry

$$D_{\text{top}} = 220 \text{ mm}, \quad D_{\text{bottom}} = 40 \text{ mm}, \quad H = 60 \text{ mm}$$

$$R_{\text{top}} = 110 \text{ mm}, \quad R_{\text{bottom}} = 20 \text{ mm}$$

Cone angle

The cone half-angle relative to the vertical axis is:

$$\theta = \tan^{-1} \left(\frac{R_{\text{top}} - R_{\text{bottom}}}{H} \right) = \tan^{-1} \left(\frac{110 - 20}{60} \right) = \tan^{-1}(1.5) \approx 56.3^\circ$$

The complementary angle relative to the ring plane is:

$$\phi = 90^\circ - \theta \approx 33.7^\circ$$

Top ring design

- Inner diameter:

$$D_{\text{top}} = 220 \text{ mm}$$

- Slot angle (relative to ring plane):

$$\phi \approx 33.7^\circ$$

- Slot direction:

– angled **inward** toward the cone axis

Bottom ring design

- Inner diameter:

$$D_{\text{bottom}} = 40 \text{ mm}$$

- Slot angle (relative to ring plane):

$$\phi \approx 33.7^\circ$$

- Slot direction:

– angled **outward** away from the cone axis

Axial spacing

The vertical distance between rings must be:

$$H = 60 \text{ mm}$$

Final design constraints

$D_{\text{top}} = 220 \text{ mm}$
$D_{\text{bottom}} = 40 \text{ mm}$
$H = 60 \text{ mm}$
$\phi \approx 33.7^\circ$

Practical fabrication note

In practice, use:

$$\phi \approx 34^\circ$$

to simplify manufacturing, and allow small tolerance for film insertion and tensioning.